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Your Body Has Its Own Built-In Ozempic

Popular weight-loss and diabetes drugs, such as Ozempic and Wegovy, target metabolic pathways that gut microbes and food molecules already play a key role in regulating

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Nutrition ▾

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[Wegovy](#), [Ozempic](#) and [Mounjaro](#) are weight loss and diabetes drugs that have made quite a splash in [health news](#). They target regulatory pathways involved in both [obesity](#) and [diabetes](#) and are widely considered breakthroughs for weight loss and blood sugar control.

But do these drugs point toward a root cause of metabolic disease? [What inspired](#) their development in the first place?

It turns out your body produces natural versions of these drugs – also known as [incretin hormones](#) – in your gut. It may not be surprising that [nutrients in food](#) help regulate these hormones. But it may intrigue you to know that the trillions of [microbes in your gut](#) are key for orchestrating this process.

I am a [gastroenterologist](#) at the University of Washington who studies how [food](#) and [your gut microbiome](#) affect health and disease. Here’s an inside-out perspective on the role natural gut hormones and healthy food play in metabolism and weight loss.

A BROKEN GUT

Specialized bacteria in your lower gut take the components of food you can’t digest like [fiber](#) and [polyphenols](#) – the elements of plants that are removed in many processed foods – and transforms them into molecules that stimulate hormones to control your [appetite](#) and [metabolism](#). These include GLP-1, a natural version of Wegovy and Ozempic.

GLP-1 and other hormones like PYY help regulate blood sugar through the pancreas. They also tell your brain that you’ve had enough to eat and your stomach and intestines to slow the movement of food along the digestive tract to allow for digestion. This system even has a name: the [colonic brake](#).

Prior to modern processed foods, metabolic regulatory pathways were under the direction of a [diverse healthy gut microbiome](#) that used these hormones to naturally regulate your metabolism and appetite. However, [food processing](#), aimed at improving shelf stability and enhancing taste, removes the bioactive molecules like [fiber](#) and [polyphenols](#) that help regulate this system.

Removal of these key food components and the resulting decrease in gut microbiome diversity may be an important factor contributing to the rise in [obesity](#) and [diabetes](#).